Natural Deduction 2

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Recap

Last week, we studied natural deduction systems:

- Intuitionistic, classical and minimal versions
- Structural properties like admissible weakening: $\Gamma \vdash A$ with $\Gamma \subseteq \Delta$ implies $\Delta \vdash A$
- Equivalence to sequent calculus: $\Gamma \Rightarrow \Delta$ if and only if $\Gamma \vdash \bigvee \Delta$
- Double-negation translation: $\Gamma \vdash_c A$ if and only if $\Gamma^* \vdash A^*$

All these results were syntactic, today we take a more semantic perspective!

Plan for Today

- Henkin's completeness proof:
  - Sketch of the usual proof
  - Constructive refinement
- Kripke semantics:
  - Version for the full syntax
  - Version for the negative fragment
- Kripke completeness:
  - Sketch of the usual proof
  - Semantic cut-elimination

Mostly based on this paper:
`https://www.ps.uni-saarland.de/Publications/documents/ForsterEtAl_2021_Completeness.pdf`

Tarski Semantics

A model $\mathcal{M}$ is a domain D together with:

- Functions $f^{\mathcal{M}} : D^{|f|} \to D$ for every function symbol f
- Predicates $P^{\mathcal{M}} \subseteq D^{|f|}$ for every predicate symbol P

Given a model $\mathcal{M}$ and environment $\rho : \mathcal{V} \to D$ we define term $\hat{\rho}(t)$ evaluation by

$$\hat{\rho}(x) := \rho(n) \qquad \hat{\rho}(f(t_1, \ldots, t_k)) := f^{\mathcal{M}}(\hat{\rho}(t_1), \ldots, \hat{\rho}(t_k))$$

and formula satisfaction $\mathcal{M} \vDash_\rho A$ by $\mathcal{M} \vDash_\rho P(t_1, \ldots, t_k) := P^{\mathcal{M}}(\hat{\rho}(t_1), \ldots, \hat{\rho}(t_k))$ and

$$\begin{aligned}
\mathcal{M} \vDash_\rho \bot &:= \bot & \mathcal{M} \vDash_\rho A \to B &:= (\mathcal{M} \vDash_\rho A) \to (\mathcal{M} \vDash_\rho B) \\
\mathcal{M} \vDash_\rho A \wedge B &:= (\mathcal{M} \vDash_\rho A) \wedge (\mathcal{M} \vDash_\rho B) & \mathcal{M} \vDash_\rho A \vee B &:= (\mathcal{M} \vDash_\rho A) \vee (\mathcal{M} \vDash_\rho B) \\
\mathcal{M} \vDash_\rho \forall x A &:= \forall d : D.\, \mathcal{M} \vDash_{\rho[x:=d]} A & \mathcal{M} \vDash_\rho \exists x A &:= \exists d : D.\, \mathcal{M} \vDash_{\rho[x:=d]} A
\end{aligned}$$

If $\mathcal{M} \vDash_\rho A$ for all $\mathcal{M}$ and ρ, we call A valid and write $\vDash A$.

Henkin Completeness Proof: Blueprint

Theorem

For every sentence A, if $\vDash A$ then $\vdash_c A$.

Outline.

1. Assume $\vDash A$ but suppose $\nvdash_c A$, so in particular $\neg A \nvdash_c \bot$.
2. Extend $\{\neg A\}$ into maximal complete theory $\mathcal{T}$ (Lindenbaum Lemma).
3. Give $\mathcal{T}$ the structure of a syntactic model $\mathcal{M}_\mathcal{T}$.
4. Establish that $B \in \mathcal{T}$ exactly if $\mathcal{M}_\mathcal{T} \vDash B$ (Truth Lemma).
5. Conclude the contradiction that $\{\neg A\}$ has a model (Model Existence). □

We go through these steps in detail, focusing on the negative $(\bot, \to, \forall)$-fragment.

Henkin Completeness Proof: Model Existence

Lemma (Lindenbaum)

For every consistent set Γ of sentences there is a consistent theory $\mathcal{T} \supseteq \Gamma$ with:

- $A \to B \in \mathcal{T}$ *if and only if* $A \in \mathcal{T}$ *implies* $B \in \mathcal{T}$
- $\forall x A \in \mathcal{T}$ *if and only if for all terms* t *we have* $A[t/x] \in \mathcal{T}$

Theorem

For every consistent set Γ of sentences there is a model $\mathcal{M}$ with $\mathcal{M} \models \Gamma$.

Proof.

1. Use the Lindenbaum Lemma to obtain $\mathcal{T} \supseteq \Gamma$ as above.
2. Define the model $\mathcal{M}_\mathcal{T}$ over the domain of (open) terms by:

$$f^{\mathcal{M}_\mathcal{T}}(t_1, \ldots, t_k) := f(t_1, \ldots, t_k) \qquad P^{\mathcal{M}_\mathcal{T}}(t_1, \ldots, t_k) := P(t_1, \ldots, t_k) \in \mathcal{T}$$

3. Establish $A[\sigma] \in \mathcal{T}$ exactly if $\mathcal{M}_\mathcal{T} \vDash_\sigma A$, so in particular $\mathcal{M}_\mathcal{T} \models \Gamma$. □

Henkin Completeness Proof: Lindenbaum Lemma

Lemma (Lindenbaum)

For every consistent set Γ of sentences there is a consistent theory $\mathcal{T} \supseteq \Gamma$ with:

- $A \to B \in \mathcal{T}$ *if and only if* $A \in \mathcal{T}$ *implies* $B \in \mathcal{T}$
- $\forall x A \in \mathcal{T}$ *if and only if for all terms* t *we have* $A[t/x] \in \mathcal{T}$

Proof.

Using an enumeration of formulas such that x_n is fresh for all A_m with $m \leq n$ we define:

$$\mathcal{T}_0 := \Gamma \cup \{A_n(x_n) \to \forall x A_n(x) \mid n \in \mathbb{N}\} \qquad \mathcal{T}_{n+1} := \mathcal{T}_n \cup \{A_n \mid \mathcal{T}_n, A \nvdash \bot\} \qquad \mathcal{T} := \bigcup_{n \in \mathbb{N}} \mathcal{T}_n$$

1. Adding Henkin axioms for $\mathcal{T}_0$ maintains consistency by use of the Drinker Paradox.
2. Each iteration step for $\mathcal{T}_{n+1}$ maintains consistency.
3. The limit $\mathcal{T}$ is maximal and thus deductively closed.
4. The wished properties follow by deductive closure and the Henkin axioms. □

Henkin Completeness Proof: Minimal Refinement

Note that for the completeness result all steps but the very first were constructive!
If we now further restrict to the $(\to, \forall)$-fragment, something interesting happens:

Lemma (Lindenbaum)

For every set Γ of sentences and every sentence $A_\bot$ there is a theory $\mathcal{T} \supseteq \Gamma$ with:

- $A_\bot \in \mathcal{T}$ *if and only if* $\Gamma \vdash_c A_\bot$
- $A \to B \in \mathcal{T}$ *if and only if* $A \in \mathcal{T}$ *implies* $B \in \mathcal{T}$
- $\forall x A \in \mathcal{T}$ *if and only if for all terms t we have* $A[t/x] \in \mathcal{T}$

Proof.

Similar as before, extending $\mathcal{T}_0 := \Gamma \cup \{A_\bot \to A_n \mid n \in \mathbb{N}\} \cup \{A(x_n) \to \forall x A_n(x) \mid n \in \mathbb{N}\}$.
For consistency, note that $\Gamma, A_\bot \to A_n \vdash_c A_\bot$ implies $\Gamma \vdash_c A_\bot$ (Peirce's Law). □

So the induced model $\mathcal{M}_\mathcal{T}$ of Γ satisfies that $\mathcal{M}_\mathcal{T} \vDash A_\bot$ implies $\Gamma \vdash_c A_\bot$.
$\Rightarrow$ From this, the completeness theorem follows constructively!

Henkin Completeness Proof: Constructive Refinement

The previous proof does not extend to the $(\bot, \to, \forall)$-fragment: If $A_\bot$ is provable, then $\mathcal{T}$ contains all formulas, including contradictions and hence doesn't induce a model.

However, we can relax the notion of model to accommodate this additional case:

Definition

We relativise the satisfaction relation to an arbitrary proposition P via $\mathcal{M} \vDash^P_\alpha \bot := P$ and call $\mathcal{M}$ exploding if for some choice of P we have $\mathcal{M} \vDash^P_\alpha \bot \to A$ for all A.

Then completeness can be derived similarly to the minimal case:

Theorem

For every sentence A, if A is valid in every exploding model then $\vdash_c A$.

Proof.

Obtain $\mathcal{T}$ as before and observe that $\mathcal{M}_\mathcal{T}$ is exploding for the choice $P := (\vdash_c A)$. □

Kripke Semantics

A standard semantics of intuitionistic logic is given by Kripke models:

Definition

A Kripke model $\mathcal{K} = (W, \preceq, (M_w)_{w \in W})$ consists of the following data:

- $(W, \preceq)$ is a pre-order, so reflexive and transitive.
- $(M_w)_{w \in W}$ is a family of models such that whenever $w \preceq w'$ and $x_1, ..., x_k \in M_w$:

$$x_1, ..., x_k \in M_{w'} \quad f^{M_w}(x_1, ..., x_k) = f^{M_{w'}}(x_1, ..., x_k) \quad P^{M_w}(x_1, ..., x_k) \to P^{M_{w'}}(x_1, ..., x_k)$$

We define the forcing relation $\mathcal{K} \Vdash^w_\alpha A$ mostly as in $M_w \vDash_\alpha A$ but with:

$$\mathcal{K} \Vdash^w_\alpha A \to B \text{ iff for for all } w' \succeq w \text{ we have that } \mathcal{K} \Vdash^{w'}_\alpha A \text{ implies } \mathcal{K} \Vdash^{w'}_\alpha B$$

$$\mathcal{K} \Vdash^w_\alpha \forall x A \text{ iff for for all } w' \succeq w \text{ and } a \in M_{w'} \text{ we have that } \mathcal{K} \Vdash^{w'}_{\alpha[x:=a]} A$$

Kripke Semantics: Soundness

Lemma (Persistence)

If $\mathcal{K} \Vdash^w_\alpha A$ *and* $w' \succeq w$*, then also* $\mathcal{K} \Vdash^{w'}_\alpha A$*.*

Theorem

If $\Gamma \vdash A$ *then* $\mathcal{K} \Vdash^w_\alpha \Gamma$ *implies* $\mathcal{K} \Vdash^w_\alpha A$ *for all* w *and* α*.*

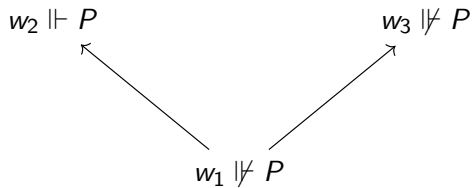
Proof.

By induction on the derivation of $\Gamma \vdash A$, here are some interesting cases:

- For the rule $\to$I, we assume that $\mathcal{K} \Vdash^w_\alpha \Gamma$ and show $\mathcal{K} \Vdash^w_\alpha A \to B$, so that for every $w' \succeq w$ we have that $\mathcal{K} \Vdash^{w'}_\alpha A$ implies $\mathcal{K} \Vdash^{w'}_\alpha B$. By persistence we hence have $\mathcal{K} \Vdash^{w'}_\alpha \Gamma, A$ and conclude $\mathcal{K} \Vdash^{w'}_\alpha B$ by the inductive hypothesis for $w := w'$.
- For the rule $\forall$I, we assume that $\mathcal{K} \Vdash^w_\alpha \Gamma$ and show $\mathcal{K} \Vdash^w_\alpha \forall x A$, so that for every $w' \succeq w$ and $a \in M_{w'}$ we have that $\mathcal{K} \Vdash^{w'}_{\alpha[x:=a]} A$. Using persistence and $x \notin FV(\Gamma)$, the first assumption also yields $\mathcal{K} \Vdash^{w'}_{\alpha[x:=a]} \Gamma$, from which we conclude $\mathcal{K} \Vdash^{w'}_{\alpha[x:=a]} A$ by the inductive hypothesis for $w := w'$ and $\alpha := \alpha[x := a]$. □

Kripke Semantics: Applications

We can give a simple Kripke models to argue non-derivabilities:



- For a language with a propositional constant P consider the model depicted
- We have $w_2 \Vdash P$ and $w_3 \Vdash \neg P$
- However, we have neither $w_1 \Vdash P$ nor $w_1 \Vdash \neg P$, so $w_1 \nVdash P \vee \neg P$
- Thus $\nvdash P \vee \neg P$ because otherwise $w_1 \nVdash P \vee \neg P$ by soundness

Kripke Semantics: Negative Fragment

To study completeness, we again restrict to the negative $(\bot, \to, \forall)$-fragment.
This allows us to simplify the semantics in the following way:

Definition

A constant domain Kripke model $\mathcal{K} = (W, \preceq, D, (M_w)_{w \in W})$ consists of the following:

- $(W, \preceq)$ is a pre-order, so reflexive and transitive.
- $(M_w)_{w \in W}$ is a family of D-models such that whenever $w \preceq w'$ and $x_1, ..., x_k \in D$:

$$f^{M_w}(x_1, ..., x_k) = f^{M_{w'}}(x_1, ..., x_k) \qquad P^{M_w}(x_1, ..., x_k) \to P^{M_{w'}}(x_1, ..., x_k)$$

We define the forcing relation $\mathcal{K} \Vdash^w_\alpha A$ mostly as before but with:

$$\mathcal{K} \Vdash^w_\alpha \forall x A \text{ iff for all } a \in D \text{ we have that } \mathcal{K} \Vdash^w_{\alpha[x:=a]} A$$

Kripke Completeness: Blueprint

Theorem

For every sentence A, if $\Vdash A$ then $\vdash A$.

Outline.

1. Assume $\vDash A$ but set out to prove $\vdash A$.
2. Consider the pre-order of consistent contexts ordered by inclusion.
3. Form the universal model $\mathcal{U}$ over the constant domain of terms by:

$$f^{M_\Gamma}(x_1, ..., x_k) = f(x_1, ..., x_k) \qquad P^{M_\Gamma}(x_1, ..., x_k) := \Gamma \vdash P(x_1, ..., x_k)$$

4. Observe that $\mathcal{U} \Vdash^\Gamma B$ exactly if $\Gamma \vdash B$ (Truth Lemma).
5. By assumption of $\vDash A$ in particular $\mathcal{U} \Vdash^\emptyset_\sigma A$, so $\vdash A$ follows. □

Kripke Completeness: Truth Lemma

Lemma (Truth Lemma)

In the universal model, we have $\mathcal{U} \Vdash^{\Gamma}_{\sigma} A$ *exactly if* $\Gamma \vdash A[\sigma]$.

Proof.

By induction on A, the only really interesting case is $A \to B$ from left to right:

1. By IH, we can assume that $\Delta \vdash A[\sigma]$ implies $\Delta \vdash B[\sigma]$ for all consistent $\Delta \supseteq \Gamma$.
2. To show $\Gamma \vdash A[\sigma] \to B[\sigma]$, it suffices to show $\Gamma, A[\sigma] \vdash B[\sigma]$.
   a. If $\Gamma, A[\sigma]$ is consistent, conclude $\Gamma, A[\sigma] \vdash B[\sigma]$ by the assumption in (1).
   b. If $\Gamma, A[\sigma]$ is inconsistent, conclude $\Gamma, A[\sigma] \vdash B[\sigma]$ by the explosion rule. □

Note that we used classical logic to test whether extension maintains consistency...

Kripke Completeness: Semantic Cut-Elimination

As in the Henkin completeness proof, we can drop the consistency requirement by relaxing to exploding models, allowing a universal model $\mathcal{U}$ over all contexts:

- Then the truth lemma is constructive as no consistency checks are necessary.
- We then can even do more, writing $\Gamma \Rightarrow A$ for cut-free sequent derivations:

$$\mathcal{U} \Vdash^{\Gamma}_{\sigma} A \text{ implies } \Gamma \Rightarrow A[\sigma]$$

  can be proven if we simultaneously prove:

$$\text{If for all } \Delta \supseteq \Gamma \text{ and } B \text{ with } \Delta, A[\sigma] \Rightarrow B \text{ we have } \Delta \Rightarrow B, \text{ then } \mathcal{U} \Vdash^{\Gamma}_{\sigma} A$$

- So by composition with soundness we in particular obtain $\Rightarrow A$ from $\vdash A$, so we eliminated cuts and since the proof is constructive it bears an actual algorithm!